

XR Avatar Prototype for Art Performance Supporting the Inclusion of Neurodiverse Artists

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Abstract

Utilizing volumetric video capture technology and augmented reality, we have developed an extended reality prototype to support the development of a hybrid Japanese Noh performance involving neurodiverse sound artists and a professional Noh singer. Through augmented reality, the volumetric video capture of the Noh performer is integrated into the live production to accommodate the needs of the neurodiverse artists. The recorded performer augmented within a physical diorama of a Noh theatre stage is used flexibly as a surrogate virtual avatar by the sound artists in developing, rehearsing, and performing. Through the artists' reflections we offer their preliminary insights in process of conceiving a performance that is mediated through extended reality tools.

CCS Concepts

• **Applied Computing** → Performing Arts.; • **Human-centered computing** → Virtual Reality.

Keywords

Virtual Reality, Augmented Reality, Disability, Sonic Arts, Volumetric Video Capture

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1 Introduction

Our research is focused on establishing avenues for collaboration between individuals with and without disabilities in the arts, using

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Figure 1: Volumetric video capture of Noh singer



Figure 2: AR occlusion test with Noh stage diorama

interactive digital technologies as a platform [Duckworth et al., 2020]. Through workshops, performances, and exhibitions, we aim to create inclusive opportunities for people of diverse abilities enabling them to co-create art and harness the inherent potential of digital technologies.

We demonstrate the implementation of an extended reality (XR) prototype developed for a live sound art performance to support the inclusion of a Melbourne based ensemble of neurodiverse artists called The Amplified Elephants [Hullick, 2013]. We reflect upon the experiences encountered by the participating artists using an XR avatar in their creative process.

2 Implementation of Avatar

We have developed XR prototypes for the participation of various people in arts activities, including people with neurodiversity. To reduce barriers to participation as well as to improve the quality of activities, we argue that it is essential to support people to become more immersed in the artistic theme in which they are working.

As a catalyst for our work, we focused on creating a contemporary interpretation of Hagoromo, a traditional Japanese tale presented through Noh theatre performance [Konparu, 2005]. To tell the Hagoromo story we embraced volumetric video capture technology to explore how virtual or augmented experiences and physical tangible props help to facilitate an understanding of the artistic theme and cultural context through intuitive and immersive tools for a diverse range of participants. We used standard virtual production methods to capture the Noh artist's performance by utilizing three Microsoft Kinect Azure™ depth cameras and DepthKit™ software to process the three volumetric data streams. We used augmented reality (AR), which allowed participants to view a 3D avatar of a Noh performer in relation to their real-world space. We developed a Unity™ application for the volumetric video playback, Vuforia for AR tracking, and customized visual effects to enhance the overall experience and narrative.

We conducted a one-day workshop with The Amplified Elephants in Tokyo to learn about traditional Noh singing and dancing. To engage in the artform, the user holds an iPad and points its camera toward a diorama, a physical scale model of a traditional Noh theatre stage approximately 40cm cubed in volume. The iPad user can select from five different scenes of the performance using the iPad touch screen. Once selected, the diorama is augmented with a virtual avatar performing a selected scene from Hagoromo. The user can move around the space and experience the augmented diorama from different perspectives. Audio of the Hagoromo recordings for each scene is provided by stereo headphones or loudspeaker. An external PC monitor allows other observers to see the same visuals as the iPad user.

The iPad app tracks the location of the diorama and sends tracking data and camera video to the PC via a dedicated Wi-Fi network. The PC renders the virtual avatar and composites the result onto the camera video and streams the video and audio back to the iPad. This combination of AR avatar and the physical diorama was used to extend new cultural understanding and artistic activities via physical and digital interaction.

3 Artists Reflections

In an interview, Ryoko Aoki, a Noh performer, provided valuable insights into the XR workshop. She found the experience of viewing her own avatar in AR particularly intriguing, as it allowed for an objective self-assessment of her performance. Additionally, she described the experience as magical, characterized by immersion and inspiration. Aoki has previously engaged in real-time, remotely delivered online concerts during the COVID-19 pandemic and recognizes the potential for continued expansion of such activities using XR technologies even post-pandemic. She expressed a desire to utilize this platform for artistic endeavors, emphasizing its unique ability to foster communication with both disabled and non-disabled individuals through advanced technology.

James Hullick, a mentor to The Amplified Elephants, provided written reflections based on observations and responses from the AR workshop. His reflections were prompted by four key questions, which sought to understand: (a) the artists' experience with XR technology, (b) their collaboration with the Noh artist using XR, (c) the benefits of these technologies for performances, and (d) their future aspirations for using the technology. The findings revealed that XR technology greatly benefited neurodiverse artists, such as The Amplified Elephants, by integrating the ancient art of Noh with modern synthesizer technology. XR technology not only eliminated physical challenges but also enabled artists to reimagine the dynamics of control within the performance, thereby enhancing their voice, agency, and interaction with the Tennin character portrayed by Ryoko Aoki. This immersive technology allowed for a collaborative environment where the neurodiverse artists felt empowered and responsible for shaping the time and emotional flow of the work.

The use of XR technology provided a significant advantage for the musicians, as it allowed the recording of the celestial dancer character, enabling performances without the physical presence of the Noh artist. This flexibility is crucial for The Amplified Elephants, who continually create new shows and do not wish their rehearsals disrupted by the scheduled availability of other artists. Additionally, the artists expressed a desire to be transformed into avatars for future collaborations with the Noh artist, ensuring the production's longevity.

4 Conclusion

XR technology offers the potential to create sensory experiences that foster empathy, allowing audiences to understand and appreciate the varied ways in which different people perceive and imagine the world. The physical diorama of the Noh stage combined with the virtual avatar provides the artists with a cultural appreciation of the traditional artform and creates a tangible bridge to communicate how the Noh artist performs on stage. This approach highlights the potential of XR to transcend traditional barriers in cross-ability and cross-cultural arts making, enabling the artists to collaborate with a virtual surrogate of the performer.

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